

02. SYSTEM ARCHITECTURE & DESIGN SPECIFICATION

Team DataDynamos | Scale Target: 100 Million Pages / Year

1. End-to-End Pipeline Architecture

The DataDynamos platform uses an asynchronous 7-stage pipeline where each stage is decoupled, timed, and monitored. Heavy OpenCV, OCR, and LLM network tasks run off the FastAPI event loop via `asyncio.to_thread` worker pools.

Stage	Technology	Functionality & Specification
Stage 1: Pre-scan	OpenCV 4.x, PIL	Automatic page deskew ($\pm 25^\circ$), 200 DPI normalization, blur & contrast check.
Stage 2: Classifier	Layout Classifier	Auto-detects document tier (CMS-1500, UB-04, Unstructured Claim, Invoice, Contract).
Stage 3: OCR Engine	Multi-Engine Swappable	PaddleOCR (PP-OCRv4 CPU), PyTesseract (v5.3), Docling (Tables), Qwen3-VL (VLM).
Stage 4: Structurer	LangExtract Framework	Serializes OCR output to structured JSON with bounding boxes & feeds OpenRouter LLMs.
Stage 5: Rule Audit	Python Rule Engine	NPI Luhn checksum (80840), ICD-10, CPT codes, UB-04 revenue charges math balance.
Stage 6: Decision	Reconciliation Agent	Merges deterministic rule guardrails + LLM judgment into final verdict.
Stage 7: HITL Loop	Prompt Memory Injector	Saves operator corrections to <code>data/feedback_memory.json</code> for zero-retraining learning.

2. OCR & Vision AI Strategy

- **PaddleOCR (PP-OCRv4) & PyTesseract (v5.3):** 97% of standard machine-printed claim forms run on ultra-fast CPU OCR engines (\$0.0001–\$0.0002/pg) with 100% local privacy.
- **Docling Layout Parsing:** Used for multi-column documents and complex tabular layouts, outputting structured Markdown tables.
- **Qwen3-VL-235B VLM Escalation:** High-cost Vision AI (\$0.0030/pg) is invoked conditionally only when block OCR confidence falls below 80%.

3. LLM Field Extraction & Spatial Grounding Layer

Instead of feeding unstructured raw text, the platform constructs a structured JSON payload containing document metadata, page blocks with exact bounding box coordinates `[x0, y0, x1, y1]`, and Markdown tables. LangExtract feeds this JSON into OpenRouter LLMs (DeepSeek-v4, GPT-4o, Claude 3.5 Sonnet), ensuring every extracted field grounds back to exact document coordinates.

4. Business Rules & Deterministic Audit Engine

- **Attending / Billing NPI Checksum:** Validates 10-digit NPIs using the National Provider Luhn algorithm (80840 US prefix).
- **ICD-10-CM & CPT Audit:** Validates clinical diagnosis and procedure code formatting.
- **UB-04 Revenue Charges Balance:** Verifies $\text{\$sum}(\text{Revenue Lines}) = \text{Total Charges}\text{\$}$.
- **Duplicate Invoice Safeguard:** Queries historical database to block double processing of identical invoice numbers.

5. Scalability for 100M+ Pages/Year & Exception Handling

- **Horizontal Scalability:** Stateless FastAPI worker nodes deployed behind load balancers with worker process pools.
- **Timeout Guardrails:** Stage-level timeouts (120s prescan, 600s OCR, 120s LLM) ensure hung requests return clean 504 errors.
- **Graceful Fallbacks:** OCR engine failures fall back to Docling; LLM extraction errors fall back to mock extraction with warning flags.